

6040 Carbonizer Media Overview

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Re: Media Overview 6040 Carbonizer

I. Manufacturers Key Points

1. Manufacturer: Tigercat Industries Inc.
2. Machine Model: 6040 Carbonizer
3. Process Description: Carbonizing Process
4. Applications: Mobile/off Road
5. Process Rate: 15-20+ tons per hour in-feed rate
6. Product manufactured: Organic carbon (Biochar)
7. Volume Reduction: 90%
8. Carbon Sequestration: 25-30%
9. Weight: 37 650 kg (83,000 lb)
10. Diesel Fuel Usage (Tier 4 final): 3.2 gallons per hour (GPH).
11. Quenching Water Usage: 150-200 gallons per hour (GPH).

II. About Tigercat Industries

1. Tigercat Industries specializes in the design and manufacture of premium quality forest harvesting systems, specialized off-road industrial machines and material processing equipment.
2. About Tigercat Link: <https://www.tigercat.com/about-sc/>

III. Media Links:

1. Tigercat Industries Website: <https://www.tigercat.com/>
2. 6040 Carbonizer Overview: <https://www.tigercat.com/product/6040-carbonizer/>
3. Carbonizing Process: <https://www.tigercat.com/feature/carbonizing-process/>



The Tigercat 6040 Carbonizer

IV. Description Of the 6040 Carbonizing Process

As the global economy shifts toward greener technologies to replace traditional methods, the 6040 carbonizer supports this transition with a carbon-negative process. In addition, the process of manufacturing organic carbon (biochar) sequesters approximately 25-30% of the wood debris' potential CO₂e back into soil or other environmental applications where it can remain for thousands of years.

The Tigercat 6040 carbonizer is a mobile, off-road processor designed to process low- and no-value clean wood debris at its point of origin. In a single step, the wood debris mass can be reduced by approximately 90%. This process utilizes the wood debris as the only fuel source for the carbonizing process. There is no other source of combustion for the process aside from the wood debris. The on-board engine provides power for the individual functions, track-drives, fans, augers, conveyor, and instrument arm. The product produced, organic carbon (biochar), is a carbon rich and porous material, which can either be worked back into the soil for beneficial gain and carbon storage or sold on the open market in place of other products such as activated carbon.

The 6040 carbonizer utilizes an over-air supply, providing a "cap" over the entire top of the carbonizing chamber, like that of an Air Curtain Burner. This technology has been proven to reduce gaseous and particulate emissions during burning. The over-air supply control device was selected for its effectiveness and for reducing both the mechanical and environmental burden associated with pre-processing wood debris.

A secondary air supply, located at the base of the machine, produces static pressure that defines a high-temperature, turbulent combustion zone at the top of the machine, just beneath the over-air supply. This high temperature and turbulent secondary combustion zone provides a residence time for volatiles to achieve total burn-out before being exhausted opposite the manifold along the length of the machine, in a defined exhaust zone.

This secondary combustion zone generates heat, which is retained in the machine's thermal ceramic sidewalls, keeping an elevated (1090°C/2000°F) and constant temperature throughout the entire primary chamber for fast and efficient operations.

The primary combustion zone, located at the base of the machine above the grates, provides a residence area with consistently high temperatures and depleted oxygen, creating the conditions necessary for pyrolysis.

As the organic carbon detaches or drops from the parent material, the char is both gravity and mechanically forced to the base of the machine, into the chamber beneath the grates. The organic carbon is immediately quenched in a fully contained and automated internal water bath, then mechanically fed to the rear of the machine, discharged onto a stacking conveyor, fed into the customers' non-combustible storage bin of choice as a final or finished product.

The carbonizer has a pass-through or processing rate of approximately 15 tons per hour with no residual left behind. The carbonizer's process results in complete in-feed pass-through. When operations are stopped, there is no fire risk, as the material is submerged in the system's sealed internal water bath prior to discharge. Typically, the product has a moisture content of 60 to 80% when conveyed into the selected storage container.

The equipment easily fits within the categories of non-road equipment that are subject to regulation under Clean Air Act Title II. Certificates of Conformity have been obtained from EPA and associated Executive Orders from the California Air Resources Board attesting to the fact that this equipment is regulated as a mobile source using compliant Tier 4 engines to self-propel the equipment on tracks over a variety of terrain.

This non-road equipment is also specifically designed to produce a useful product, organic carbon (biochar), which can be used as a soil amendment on site or recovered to be sold commercially. This is a unique feature of the equipment and accomplished through several design features:

- High temperature pyrolysis of wood, producing a thermal decomposition of source materials. The equipment typically operates in a range of 1090°C/2000°F.
- Incorporation of an automated mechanical system to first gather, then transport and dispense biochar during operation of the unit.
- Use of a continual quenching system (steam activation) to introduce moisture into the char, halt the pyrolysis process and increase the porosity of the product.

The net result of utilizing this equipment for addressing wood and agricultural debris is that approximately 25 to 30% of the carbon from the wood waste is captured. This produces a product that ranges from 90 to 95% carbon content. As mentioned above, biochar can and is often used as a soil enhancer or soil amendment. But the equipment also effectively displaces the use of other fossil fuels that are associated with alternative methods to address wood and agricultural waste, e.g., the need to transport resulting ash to landfills and/or move wood and agricultural waste to fixed-site facilities.

To limit global warming, reducing emissions and removing significant amounts of CO₂ from the atmosphere is crucial. The carbon-negative process of the 6040 carbonizer actively combats the accumulation of greenhouse gases contributing to climate change. Implementing carbon-negative solutions faces challenges including scalability, cost, potential environmental impacts, and ensuring the captured carbon is stored permanently. The carbonizer addresses each of these difficulties and machines are currently producing carbon credits.

IV. Organic Carbon (Biochar) Description

Biochar is a specific, highly stable type of organic carbon. Tigercat Industries' 6040 Carbonizer manufactures organic carbon (biochar) by heating biomass to high temperatures in a controlled environment with limited oxygen, a carbonizing process.

This process differs significantly from simple burning or incineration, which would release the carbon into the atmosphere, producing a non-adsorbing and low carbon ash. The specific temperature and duration of the carbonizing process impacts the properties of the resulting biochar.

Benefits of biochar in soil applications

Adding biochar to soil offers multiple environmental and agricultural benefits:

Improves soil health: Its porous structure improves soil aeration, drainage, and water-holding capacity, particularly in degraded or sandy soils.

Increases fertility: Biochar enhances the soil's ability to retain nutrients by increasing its cation exchange capacity (CEC). This holds nutrients like nitrogen and phosphorus in the soil, preventing them from leaching away.

Enhances plant growth: By creating a healthier, more nutrient-rich environment, biochar can boost crop yields and promote stronger root systems.

Stimulates beneficial microbes: The countless pores and crevices provide a habitat for beneficial soil microbes, fungi, and earthworms, which further improve soil health and nutrient cycling.

Sequestering carbon: Biochar is highly stable and can store carbon in the soil for hundreds to thousands of years. This long-term carbon storage, known as carbon sequestration, helps remove carbon dioxide from the atmosphere.

Mitigates greenhouse gases: By stabilizing carbon that would otherwise decompose and release greenhouse gases like carbon dioxide and methane, biochar contributes to climate change mitigation.

Benefits of biochar in non-soil applications

Besides its use in agriculture to improve soil, biochar is a highly versatile material with a wide range of non-soil applications. Its high porosity, large surface area, and absorptive qualities make it useful for environmental remediation, in building materials, for animal health, and even in textiles and personal care products.

Environmental remediation

Water filtration: Biochar's porous structure makes it an effective and low-cost filtration medium, similar to activated carbon. It is used to remove a wide variety of contaminants, including heavy metals, pesticides, pharmaceutical compounds, and industrial pollutants from wastewater and storm water.

Controlling odors: Biochar can absorb and neutralize odors in a variety of settings. It is used to reduce unpleasant smells from compost piles, manure, wastewater treatment facilities, and urban runoff.

Methane reduction in agriculture: As a feed additive for livestock, biochar can improve digestion and reduce methane emissions from animals like cows. It also helps mitigate methane released during manure and compost management.

Building and construction

Insulation and building materials: When mixed with clay, lime, or cement, biochar can be used to create high-performance building materials. Biochar-based plasters, bricks, cement offer excellent thermal insulation, regulate humidity, and can adsorb toxins and odors from the air.

Carbon sequestration: By locking carbon into building materials, biochar creates “carbon sinks” in buildings. At the end of a building’s life, the biochar can be safely recycled into compost, extending its carbon sequestration value.

Durability and strength: Biochar can serve as a filler material in concrete and composites to reduce weight and potentially improve fracture resistance and strength, depending on the feedstock and production temperature.

Animal feed and husbandry

Feed additive: Ingesting biochar can improve the nutrient absorption and digestive health of livestock. It has been shown to neutralize toxins, reduce the incidence of diarrhea, and improve feed efficiency in farm animals.

Bedding and composting: Biochar can be added to animal bedding to absorb moisture and odors and then co-composted with manure. This process helps retain nutrients and reduces greenhouse gas emissions from manure storage.

Aquaculture: Biochar can improve water quality in aquaculture by absorbing ammonia and other pollutants. It is also used as a feed supplement for fish and other aquatic animals to enhance their growth and health.

Textiles and personal care

Odor and moisture control: Biochar is woven into fabrics for functional clothing, socks, and shoe inserts. This provides thermal regulation and helps reduce odors by absorbing moisture and sweat.

Skin and wellness products: Biochar’s absorbent properties are used in personal care items like soaps, facial masks, toothpastes, and bath additives. It helps to deep clean, remove odors, and promote

Mattresses and pillows: As a filler for bedding, biochar can regulate temperature, absorb perspiration, and help provide a more comfortable sleeping environment.